Learning stack: EFE, Dirichlet updates, and BMR

Active-inference agents do more than infer states under a fixed model: they learn the parameters of the model, score policies by expected free energy, and revise model structure.

Learning stack (part 2)

The active-inference community has standardized all three operations (Da Costa et al. 2020; Smith et al. 2022), and Friston et al. (2024) place them at the heart of the federated belief-sharing scenario.

Learning stack (part 3)

We reimplement each in closed form so the language-acquisition, expected-free-energy, and emergence studies of sec. 8.22 rest on machine-checkable quantities rather than fitted curves.

Conjugate Dirichlet learning from co-occurrence counts

A sentinel learns its observation model A by placing a Dirichlet prior with concentration a on each column and updating it conjugately from observation-state co-occurrence counts (Friston et al. (2024), their equations 9–12).

Conjugate Dirichlet learning... (part 2)

One learning step adds the expected sufficient statistics for that step and reads off the column-normalized posterior mean,

Conjugate Dirichlet learning... (part 3)

$$a \leftarrow a + \text{counts}, \qquad \mathbb{E}[A]_{os} = \frac{a_{os}}{\sum_{o'} a_{o's}}, \tag{18}$$

Conjugate Dirichlet learning... (part 4)

implemented by `learn_likelihood` in the `dirichlet_learning` module. Intuitively, each concentration vector is a running tally of how often each outcome was seen while the creature occupied a given state: the prior seeds that tally with pseudo-counts, every step adds the co-occurrences it witnessed,

Conjugate Dirichlet learning... (part 5)

and the posterior mean is simply the tally renormalized into a categorical.

Likelihood learning is therefore bookkeeping — accumulate counts, then normalize — with no iterative optimization loop.

Conjugate Dirichlet learning... (part 6)

We drive eq. 18 with the expected sufficient statistics under the true model — a fixed count batch count_scale $\cdot A^{\star}$ per step, optionally jittered by a seeded generator. As the concentrations accumulate,

Conjugate Dirichlet learning... (part 7)

the expected likelihood $\mathbb{E}[A]$ converges to the data-generating $A^\star$; convergence is measured by the per-column KL divergence summed over hidden states,

Conjugate Dirichlet learning... (part 8)

$$\mathrm{KL}(A^\star \| \mathbb{E}[A]) = \sum_s \sum_o A^\star_{os} \cdot \ln \frac{A^\star_{os}}{\mathbb{E}[A]_{os}}. \tag{19}$$

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Conjugate Dirichlet learning... (part 9)

which decreases monotonically toward zero — the standard-Bayes / KL fixed point. The learned likelihood always has full support (the Dirichlet prior is strictly positive), so eq. 19 is finite throughout. ISC-17 pins the monotone-decreasing KL trajectory,

Conjugate Dirichlet learning... (part 10)

and the language-acquisition study of sec. 8.22 reports the descent of eq. 19 across 24 steps.

Conjugate Dirichlet learning… (part 11)

The implementation also carries the η forgetting hyperprior of Friston et al. (2024), their equation 12: before each conjugate addition the running concentration is decayed so the *total* concentration mass saturates at η rather than growing without bound,

Conjugate Dirichlet learning... (part 12)

modeling an agent that stays adaptable instead of becoming infinitely confident. With η unset the classical unbounded accumulation of eq. 18 is recovered.

Expected free energy as the action-selection objective

A sentinel scores a candidate policy π by its expected free energy $G(\pi)$, which the active-inference formulation decomposes two equivalent ways (Friston et al. (2024), their equation 2):

Expected free energy as... (part 2)

a cost view of risk plus ambiguity, and a value view of pragmatic plus epistemic value.

Expected free energy as... (part 3)

The two views are the same scalar rearranged, stated as eq. 22 and pinned to a zero residual by the algebraic identity eq. 23 in sec. 9.

Expected free energy as... (part 4)

We compute every term in closed form from the categorical model (A, B, C, D_0) in `expected_free_energy.decompose`:

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Risk

Risk is $\mathrm{KL}(q(o \mid \pi) \,\|\, p_C(o))$, the deviation of the policy-predicted outcomes from the preferred-outcome pmf $p_C(o) = \mathrm{softmax}(C)[o]$. Write $q_\pi(o) := q(o \mid \pi)$ for this scored-policy outcome predictive,

Risk (part 2)

used in the remaining terms.

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Ambiguity

Ambiguity is the expected likelihood entropy $\mathbb{E}_{q(s)}\big[H\big[p(o \mid s)\big]\big]$, the outcome uncertainty given the state.

Pragmatic value

Pragmatic value is the expected log-preference $\mathbb{E}_{q_\pi(o)}[\ln p_C(o)]$ — the utility, exploitation term.

Epistemic value

Epistemic value is the state-outcome mutual information $H[q_\pi(o)] - \mathbb{E}_{q(s)}[H[p(o \mid s)]]$ — the expected information gain that drives exploration.

Epistemic value (part 2)

Because there is no sampling, the identity of eq. 23 holds to floating-point tolerance; ISC-19 (`expected_free_energy`) pins the residual of the decomposition to zero and pins each term's semantics independently (deterministic likelihoods give zero ambiguity;

Epistemic value (part 3)

uninformative likelihoods give zero epistemic value; preference-matched predictions lower risk).

Epistemic value (part 4)

fig. 11 visualizes the additive cost view and the signed pragmatic/epistemic waterfall terminating at $G(\pi)$ — a deterministic identity (Proposition 7), not a fitted result.

Bayesian model reduction for the configured pruning sign control

Sentinels also revise model *structure*. Bayesian model reduction (BMR) scores whether a reduced model — for example one that prunes a redundant location column by shrinking its concentration toward zero —

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Bayesian model reduction… (part 2)

has more evidence than the full model, *without re-running inference* (Friston & Penny via the post-hoc model optimization lineage (Friston and Penny 2011); the same Beta-function identity is their equation 13 (Friston et al. (2024))).

Bayesian model reduction... (part 3)

Because the likelihood is shared, the reduced posterior is available in closed form, $\text{reduced_post} = \text{post} + \text{reduced_prior} - \text{prior}$, and the change in (negative) variational free energy is a difference of log multivariate Beta functions,

Bayesian model reduction... (part 4)

$$\begin{aligned} \Delta F &= \ln B(\text{prior}) - \ln B(\text{post}) \\ &\quad + \ln B(\text{reduced_post}) \\ &\quad - \ln B(\text{reduced_prior}), \\ \ln B(a) &= \sum_k \ln \Gamma(a_k) - \ln \Gamma\Big(\sum_k a_k\Big). \end{aligned} \tag{20}$$

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Bayesian model reduction... (part 5)

computed in `bayesian_model_reduction.reduce`. A positive ΔF in eq. 20 means the reduced model carries more evidence — the pruned structure was redundant and should be adopted; a negative ΔF means the reduction destroyed support the data require.

Bayesian model reduction... (part 6)

When the reduced prior equals the prior the score is identically zero in exact algebra, a zero point the suite pins to machine precision (ISC-20).

Bayesian model reduction... (part 7)

The configured BMR sign-control study of sec. 8.22 uses this operation over $n = 4$ candidate states. It contrasts a redundant reduction ($\Delta F = 3.68$ nats; adopt) with a supported one ($\Delta F = -27.67$ nats; reject) in fig. 15.

Bayesian model reduction... (part 8)

This fixed-candidate algebraic comparison is deterministic, so it has no resampled sample or bootstrap interval and does not establish structure discovery or universal emergence.